MINI PROJECT REPORT

Student Record and Academic Management Platform Using Agile Methodology

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TITLE PAGE

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1. Abstract

The Student Record and Academic Management Platform is a software project designed to manage student information and academic activities in an educational institution. The system helps administrators and faculty members maintain student records, manage courses, record attendance, enter marks, calculate grades, and generate reports.

In the traditional method, student information is often maintained in registers or separate files. This makes searching, updating, and maintaining records time-consuming. The proposed system aims to make these activities easier, faster, and more organized.

The project follows the Agile software development methodology. Agile helps the development team divide the project into smaller tasks and complete them in multiple sprints. Scrum roles, user stories, acceptance criteria, story points, sprint planning, and a Scrum board are used to manage project development.

The main objective of this project is to develop a simple, secure, and user-friendly system that improves the management of student academic information.

2. Introduction

Educational institutions handle a large amount of student information every year. This information includes personal details, course details, attendance records, examination marks, grades, and academic reports.

Managing all this information manually can create problems such as duplicate records, calculation mistakes, difficulty finding information, and delays in preparing reports.

The Student Record and Academic Management Platform provides a centralized system for maintaining student information. Authorized users can add, view, update, and search student records.

The project uses Agile methodology to plan and manage the development process. Instead of developing the complete application at once, the team develops it in small parts and tests each part before moving to the next sprint.

2.1 Problem Statement

Manual student record management requires considerable time and effort. Maintaining separate registers and files can result in errors, data duplication, difficulty retrieving information, and delays in generating academic reports. Therefore, a computerized platform is required to manage student records, courses, attendance, marks, and reports in an efficient and organized manner.

2.2 Objectives

The main objectives of the project are:

1st To maintain student information in a centralized system.
2nd
3rd To provide student registration and profile management.
4th
5th To manage course details.
6th
7th To record and view student attendance.
8th
9th To enter examination marks and calculate grades.
10th
11th To generate student and academic reports.
12th
13th To provide secure login for authorized users.
14th
15th To reduce manual work and data entry errors.
16th
17th To use Agile methodology for planning, development, and testing.
18th

2.3 Scope of the Project

The project covers the following activities:

- Student registration and profile management.
- Searching, updating, and deleting student records.
- Adding and managing courses.
- Recording and viewing attendance.
- Entering marks and calculating grades.
- Generating attendance and performance reports.
- User authentication and access control.

Advanced features such as online fee payment, biometric attendance, and integration with external university systems are outside the initial project scope.

3. Business Requirements

Business requirements explain what the institution expects from the proposed system.
The main business requirements are:
1st The system should store student information in an organized manner.
2nd
3rd Faculty members should be able to maintain attendance records.
4th
5th Academic marks should be stored and updated correctly.
6th
7th Administrators should be able to manage student and course information.
8th
9th Users should be able to retrieve records quickly.
10th
11th The system should generate useful academic reports.
12th
13th Student information should be protected from unauthorized access.
14th

3.1 Functional Requirements

Functional requirements describe the operations performed by the system.

Requirement ID	Functional requirement
FR1	User registration and login
FR2	Add new student records
FR3	Search student records
FR4	Update student information
FR5	Delete student records
FR6	Add and update course details
FR7	Record student attendance
FR8	Enter and update examination marks

FR9	Calculate grades
FR10	Generate academic reports

3.2 Non-Functional Requirements

Non-functional requirements describe the quality and performance expected from the system.
1st Security: Only authorized users should access protected information.
2nd
3rd Usability: The interface should be simple and easy to understand.
4th
5th Performance: Student records should be retrieved within a reasonable time.
6th
7th Reliability: The system should save records accurately.
8th
9th Maintainability: Developers should be able to update the software easily.
10th
11th Availability: The system should be accessible when required, subject to server availability.
12th
13th Data integrity: The system should avoid duplicate or inconsistent records.
14th

4. Agile Product Vision

4.1 Product Vision Statement

“To develop a secure, simple, and user-friendly Student Record and Academic Management Platform that enables educational institutions to manage student records, courses, attendance, academic performance, and reports efficiently through a centralized software application.”

4.2 Mission Statement

The mission of the project is to reduce manual work, improve the accuracy of academic records, simplify student information management, and provide authorized users with easy access to important academic information.

4.3 Primary Users

The primary users of the system are:

- Administrator: Manages student records, courses, and user accounts.
-
- Faculty Member: Records attendance and enters examination marks.
-
- Student: Views permitted personal and academic information.
-
- Institution Management: Uses academic reports to monitor student performance.
-

4.4 Business Benefits

-
- Reduces paperwork.
-
- Saves time when searching for records.
-
- Improves accuracy of attendance and marks.
-

- Simplifies report generation.
-
- Provides organized student information.
-
- Improves coordination between administrators and faculty.
-

4.5 Success Criteria

The project will be considered successful when:
1st Users can log in using valid credentials.
2nd
3rd Administrators can add, search, update, and delete student records.
4th
5th Faculty members can record attendance and examination marks.
6th
7th The system calculates grades according to defined rules.
8th
9th Reports display correct information.
10th
11th Invalid input is handled with appropriate error messages.
12th
13th All essential functions pass their planned tests.
14th

4.6 Project Constraints

-
- Limited project development time.
-
- Limited team members.
-
- Limited technical experience.
-
- Availability of computers and development software.
-
- Need to protect student information.
-
- Requirement to complete the project within the academic schedule.
-

5. Minimum Viable Product (MVP)

A Minimum Viable Product is the first basic version of a software application that contains the most important features.
The MVP for this project includes:
1st User login.
2nd
3rd Add student.
4th
5th Search student.
6th
7th Update student.
8th
9th Delete student.
10th

11thAdd course.
12th
13thRecord attendance.
14th
15thEnter examination marks.
16th
17thGenerate basic reports.
18th
Features such as advanced analytics, automated notifications, and external system integration can be added in later releases.

6. Project Roadmap

The project roadmap explains the planned development stages.

Phase	Main activities	Expected outcome
Phase 1	Requirement analysis and project setup	Requirements and initial design
Phase 2	Login and authentication	Secure login module
Phase 3	Student management	Student record module
Phase 4	Course and attendance management	Course and attendance modules
Phase 5	Marks and reporting	Academic performance reports
Phase 6	Testing and improvements	Tested project release

7. Scrum Team Structure

Scrum is an Agile framework used to manage software development through short development cycles called sprints.
The project team consists of the following roles.

7.1 Product Owner

Responsibilities:

- Understand the needs of stakeholders.
- Define product requirements.
- Prepare and prioritize the Product Backlog.
- Decide which features are most important.
- Review completed work.
- Provide feedback to the development team.

7.2 Scrum Master

Responsibilities:

- Help the team follow Scrum practices.
- Organize Scrum meetings.
- Identify obstacles affecting development.

-
- Encourage communication and teamwork.
-
- Help the team improve its working process.
-

7.3 Development Team

Responsibilities:
- Design the application.
-
- Write program code.
-
- Develop the database.
-
- Test application features.
-
- Fix errors.
-
- Prepare project documentation.
-
- Demonstrate completed features.
-

7.4 Stakeholders

Stakeholders are people who are interested in or affected by the project.
The main stakeholders are students, faculty members, administrators, institution management, and the project development team.

8. Project Epics

An Epic is a large feature or functional area that can be divided into smaller user stories.

Epic 1: Student Management

Purpose: To manage student information.
Features:
- Student registration.
-
- Add student records.
-
- View student profiles.
-
- Search student records.
-
- Update student details.
-
- Delete student records.
-

Epic 2: Course Management

Purpose: To manage course information.
Features:
- Add new courses.

-
- Update course details.
-
- View course information.
-
- Assign courses to students.
-
- Display course lists.
-

Epic 3: Attendance Management

Purpose: To maintain student attendance records.
Features:

- Select a course.
-
- Select a lecture date.
-
- Mark students as present or absent.
-
- Save attendance.
-
- View attendance history.
-
- Generate attendance reports.
-

Epic 4: Marks and Academic Performance

Purpose: To maintain examination marks and academic results.
Features:

- Enter examination marks.
-
- Update marks.
-
- Calculate grades.
-
- View academic performance.
-
- Generate result sheets.
-

Epic 5: Authentication and Security

Purpose: To protect the system from unauthorized access.
Features:

- User login.
-
- User logout.
-
- Password reset.
-
- Role-based access control.
-
- Validation of login credentials.
-

Epic 6: Notifications

Purpose: To provide important academic updates.
Features:

- Attendance alerts.
-
- Examination notifications.
-
- Result notifications.
-

Epic 7: Reporting

Purpose: To provide useful information in report format.
Features:

- Student reports.
-
- Attendance reports.
-
- Academic performance reports.
-
- Course reports.
-

9. User Stories and Acceptance Criteria

A user story describes a feature from the point of view of the person using the system.
The standard format is:
“As a [user], I want [feature], so that [benefit].”
Acceptance criteria are the conditions that must be satisfied for a feature to be considered complete.

User Story 1: User Login

User Story: As an administrator, I want to log in to the system so that I can manage student information securely.
Acceptance Criteria:
1st The login page should open successfully.
2nd
3rd Registered users should be able to log in with valid credentials.
4th
5th Invalid credentials should display an error message.
6th
7th Successful login should open the appropriate dashboard.
8th
9th Unauthorized users should not access protected pages.
10th

User Story 2: Add Student

User Story: As an administrator, I want to add a new student so that student information is stored in the system.
Acceptance Criteria:
1st The system should display a student registration form.
2nd
3rd The user should enter the required student details.
4th
5th Required fields should not be left empty.
6th

7th Duplicate student IDs should be rejected.
8th
9th Valid student information should be saved successfully.
10th

User Story 3: Search Student

User Story: As an administrator, I want to search for a student so that I can find student information quickly.
Acceptance Criteria:
1st The system should provide a search field.
2nd
3rd The user should be able to search using a student ID or name.
4th
5th Matching student records should be displayed.
6th
7th A suitable message should appear when no record is found.
8th

User Story 4: Update Student

User Story: As an administrator, I want to update student details so that the information remains correct.
Acceptance Criteria:
1st The administrator should select an existing student.
2nd
3rd The system should display the current information.
4th
5th The administrator should be able to edit permitted fields.
6th
7th Valid changes should be saved.
8th
9th Updated information should appear when the record is viewed again.
10th

User Story 5: Record Attendance

User Story: As a faculty member, I want to record student attendance so that attendance records remain accurate and updated.
Acceptance Criteria:
1st The faculty member should be able to select a course.
2nd
3rd The faculty member should be able to select a lecture date.
4th
5th Students should be marked present or absent.
6th
7th Attendance records should be saved successfully.
8th
9th The saved attendance should be available for reporting.
10th

User Story 6: Enter Marks

User Story: As a faculty member, I want to enter examination marks so that student academic performance can be maintained.
Acceptance Criteria:
1st The faculty member should select the relevant course or examination.
2nd

3rd The system should accept marks within the defined maximum.
4th
5th Invalid marks should be rejected.
6th
7th Valid marks should be saved.
8th
9th Saved marks should be available for viewing and reporting.
10th

User Story 7: Generate Reports

User Story: As an administrator, I want to generate academic reports so that I can review student performance and attendance.
Acceptance Criteria:
1st The user should be able to select the report type.
2nd
3rd The system should retrieve the relevant records.
4th
5th The report should display accurate information.
6th
7th The system should show a suitable message if no data is available.
8th
9th Authorized users should be able to view the report.
10th

10. Product Backlog

The Product Backlog is a prioritized list of features, improvements, and tasks required to develop the project.

Priority	Epic	User story
High	Authentication	User Login
High	Student Management	Add Student
High	Student Management	Search Student
High	Student Management	Update Student
High	Attendance	Record Attendance
High	Marks Management	Enter Marks
Medium	Course Management	Add Course
Medium	Reporting	Generate Attendance Report
Medium	Reporting	Generate Performance Report
Medium	Notifications	Attendance Alert
Low	Dashboard	Academic Analytics

High-priority items are developed first because they provide the essential functions of the application. Medium- and low-priority items can be developed in later sprints according to available time and resources.

11. Story Point Estimation

Story points are used to estimate the relative effort and complexity of a user story. They do not represent exact hours.
The team can use the following estimation scale, as shown in the reference document.

Story points	Complexity

1	Very Easy
2	Easy
3	Moderate
5	Difficult
8	Very Difficult
13	Highly Complex

11.1 Story Point Estimation Sheet

User story	Story points
User Login	3
Add Student	3
Search Student	2
Update Student	3
Add Course	3
Record Attendance	5
Enter Marks	5
Generate Reports	5
Attendance Notifications	5

These are proposed estimates for the mini-project. The actual estimates should be agreed upon by the team during Sprint Planning or Planning Poker.

12. Sprint Planning

A sprint is a short development period during which the team completes a selected group of tasks.
The project is divided into three proposed sprints.

12.1 Sprint 1: Project Setup and Authentication

Sprint Goal: Set up the project and develop the login feature.
Activities:
1st Initialize the project repository.
2nd
3rd Analyze the requirements.
4th
5th Design the basic user interface.
6th
7th Develop the login functionality.
8th
9th Validate login credentials.
10th
11th Test valid and invalid login attempts.
12th
Expected Deliverables:

- Project structure.
-
- Basic user interface.
-

- Login module.
-
- Initial test results.
-

12.2 Sprint 2: Student Management

Sprint Goal: Develop the student record management module.
Activities:
1st Design the student registration form.
2nd
3rd Add student records.
4th
5th Search for students.
6th
7th Update student details.
8th
9th Delete student records.
10th
11th Test the student management functions.
12th
Expected Deliverables:

- Student registration module.
-
- Student search functionality.
-
- Update and delete functionality.
-
- Student management test results.
-

12.3 Sprint 3: Course, Attendance and Academic Records

Sprint Goal: Develop basic course and academic record management.
Activities:
1st Add course details.
2nd
3rd Assign courses to students.
4th
5th Record student attendance.
6th
7th Enter examination marks.
8th
9th Calculate grades.
10th
11th Generate basic academic reports.
12th
13th Test the completed features.
14th
Expected Deliverables:

- Course management module.
-
- Attendance management module.
-
- Marks management module.
-

- Basic academic reports.
- Final testing results.

13. Scrum Board Development

A Scrum board helps the team track the status of project tasks.
The board contains five columns.

- Backlog: All identified tasks.
- To Do: Tasks selected for development.
- In Progress: Tasks currently being developed.
- Testing: Tasks being checked for errors.
- Completed: Tasks that have met the completion criteria.

13.1 Example Scrum Board

Backlog	To Do	In Progress	Testing	Completed
Notifications	Add Student	Login Module	Student Search	Project Setup
Academic Analytics	Add Course	Attendance	Marks Entry	Requirements
Advanced Reports	Update Student	—	—	Basic Design

Note: This is an illustrative Scrum board. Tasks should be moved between columns as actual development progresses.

14. Scrum Ceremonies

Scrum ceremonies are meetings that help the team plan work, communicate progress, review results, and improve its process.

14.1 Sprint Planning

In Sprint Planning, the team selects the user stories to complete during the sprint.
Activities:

- Review the Product Backlog.
- Select high-priority user stories.
- Estimate story points.
- Define the Sprint Goal.
- Assign responsibilities.

14.2 Daily Scrum

The Daily Scrum is a short meeting used to discuss progress and obstacles.
Each team member answers three questions:
1st What did I complete yesterday?

2nd
3rd What will I work on today?
4th
5th What problems or challenges am I facing?
6th
Example:

- Yesterday, I completed the student registration form.
- Today, I will develop student search functionality.
- I am facing an issue with validating duplicate student IDs.

14.3 Sprint Review

The Sprint Review takes place at the end of the sprint.
Activities:

- Demonstrate completed features.
- Explain the work completed.
- Collect feedback from stakeholders.
- Identify changes or improvements.
- Update the Product Backlog if necessary.

14.4 Sprint Retrospective

The Sprint Retrospective helps the team improve its development process.
What went well?

- Team members communicated effectively.
- Tasks were divided properly.
- Completed features were tested.

What challenges occurred?

- Some tasks required more time than expected.
- Errors occurred during testing.
- Some requirements needed clarification.

What improvements should be made?

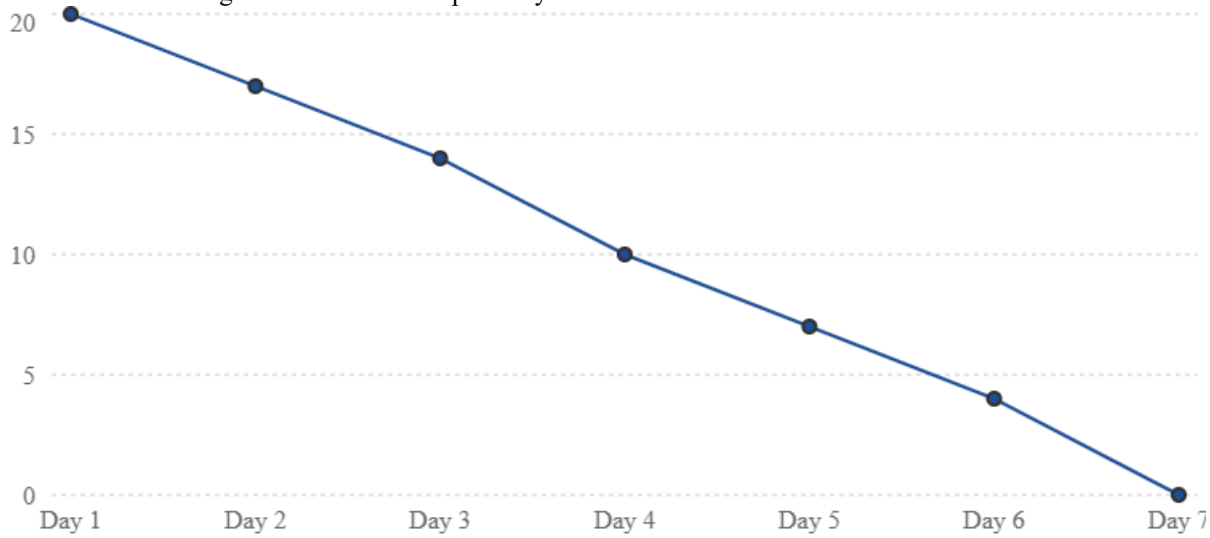
- Improve task estimation.
- Test features regularly.
- Communicate problems earlier.
- Review requirements before starting development.

15. Burndown Chart

A burndown chart shows how much work remains during a sprint.
The following is an illustrative Sprint 1 example, assuming the team begins with 20 story points of work.

Sprint day	Remaining story points
Day 1	20
Day 2	17
Day 3	14
Day 4	10
Day 5	7
Day 6	4
Day 7	0

Example Sprint Burndown
Illustrative remaining work across seven sprint days.



The chart shows the remaining work decreasing as tasks are completed. In a real project, the values should be updated using the team's actual progress.

16. Comparison of Agile Frameworks

Agile is an approach to software development that emphasizes incremental delivery, collaboration, feedback, and adapting to change.
The following frameworks are considered in the reference document.

Framework	Description	Suitability for this project
Scrum	Uses sprints, defined roles, and regular meetings	Highly suitable for structured student project development
Kanban	Uses a visual board to manage continuous workflow	Suitable for tracking tasks and reducing work in progress
Extreme Programming (XP)	Emphasizes coding practices, testing, and frequent releases	Useful when code quality and frequent testing are important
Lean Development	Focuses on reducing waste and maximizing value	Useful for avoiding unnecessary features

LeSS	Applies Scrum principles to larger development organizations	More suitable for multiple Scrum teams
SAFe	Coordinates Agile work across large organizations	Generally unnecessary for a small academic project
Scrum@Scale	Coordinates multiple Scrum teams	More suitable for projects involving several Scrum teams

16.1 Recommended Agile Framework

Scrum is recommended for this mini-project because it provides a clear structure for dividing work into sprints, assigning responsibilities, reviewing progress, and improving the software through feedback.
Kanban can also be used alongside the team's planning process to display task progress visually.

17. Testing and Quality Assurance

Testing is performed to verify that the system works according to the defined requirements.

17.1 Types of Testing

1st
2ndUnit Testing: Tests individual functions or modules.
3rd
4th Integration Testing: Checks whether different modules work together.
5th
6th Functional Testing: Verifies that features meet their requirements.
7th
8th Validation Testing: Checks that invalid data is handled correctly.
9th
10thUser Acceptance Testing: Confirms that the system meets user needs.
11th

17.2 Sample Test Cases

Test case	Input or action	Expected result
Login with valid credentials	Correct username and password	Login successful
Login with invalid credentials	Incorrect password	Error message displayed
Add student	Enter valid student details	Student record saved
Add duplicate student	Enter an existing student ID	Duplicate rejected
Search student	Enter an existing student ID	Matching record displayed
Enter invalid marks	Marks exceed maximum allowed	Validation error displayed
Record attendance	Mark attendance for a course	Attendance saved
Generate report	Select a valid report type	Relevant report displayed

18. Security and Risk Management

Security is important because the platform stores student information.

18.1 Security Measures

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- Require authentication for protected features.
- Use role-based access control.
- Validate user inputs.
- Protect passwords using appropriate password-hashing methods.
- Restrict access to student information.
- Maintain backups of important records.
- Provide appropriate error messages without exposing sensitive details.

18.2 Project Risks

Risk	Possible impact	Mitigation
Incorrect requirements	Features may not meet user needs	Review requirements with stakeholders
Development delays	Project may not finish on time	Prioritize essential features
Coding errors	Application may not work correctly	Test code regularly
Data loss	Student records may be lost	Maintain backups
Unauthorized access	Student data may be exposed	Apply authentication and access controls

19. Project Timeline

The following is a proposed four-week schedule for the mini-project.

Week	Activities
Week 1	Requirement analysis, product vision, project setup, and login
Week 2	Student registration, search, update, and delete
Week 3	Course management, attendance, and marks
Week 4	Reports, testing, corrections, and documentation

The timeline may be adjusted according to project complexity and team availability.

20. Expected Outcomes

After completing the project, the expected outcomes are:
1st A centralized system for managing student information.
2nd
3rd A simple login mechanism for authorized users.
4th
5th A student management module.
6th
7th Basic course and attendance management.
8th
9th Examination marks and grade management.
10th
11thBasic academic reporting functionality.

12th
13thExperience with Agile planning and development.
14th
15thPractical understanding of Scrum roles and ceremonies.
16th
17thExperience preparing user stories, acceptance criteria, and a Product Backlog.
18th
19thImproved teamwork, communication, and testing skills.
20th

21. Conclusion

The Student Record and Academic Management Platform is a proposed software solution for simplifying student information management in educational institutions. It provides a structured approach to handling student details, courses, attendance, examination marks, and academic reports.

The project follows Agile methodology to divide the work into smaller tasks and complete them through planned sprints. Scrum roles, user stories, acceptance criteria, story points, a Product Backlog, a Scrum board, and regular review meetings help the team organize its work effectively.

This mini-project also provides practical knowledge of software planning, teamwork, testing, and continuous improvement. By implementing the proposed system, educational institutions can reduce manual work and maintain academic information in a more organized manner.

22. References

1st
2ndUNIT-2 AGILE MINI-PROJECT, provided reference document.
3rd
4th Class notes and study materials on Agile Software Development.
5th
6th Scrum framework concepts covered in the project syllabus.
7th

Important: This report is a proposed project plan based on your reference document. It does not claim that the application has already been developed or tested. If your teacher requires screenshots, source code, database design, or actual implementation results, add those after completing the corresponding work.